

Problem 1: Sum of Two Numbers (Using a Method)

Source code:

```
package LB_Assingment3;

import java.util.Scanner;

public class Sum_of_2_num {
    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

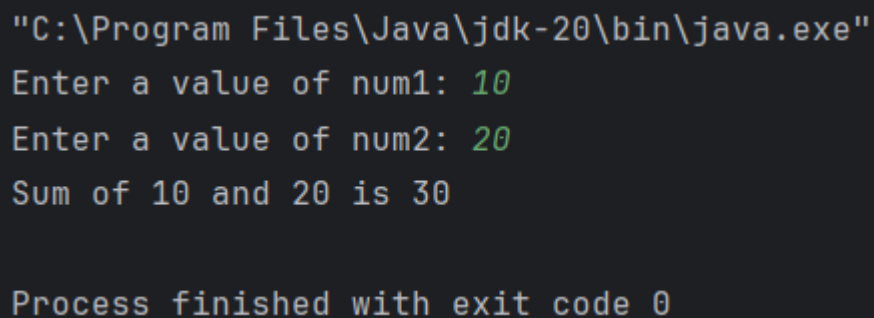
        System.out.print("Enter a value of num1: ");
        int num1=sc.nextInt();

        System.out.print("Enter a value of num2: ");
        int num2=sc.nextInt();

        int sum=SumOfTwoNumbers(num1,num2);
        System.out.println("Sum of "+num1+" and "+num2+" is "+sum);
    }
    public static int SumOfTwoNumbers(int num1,int num2){

        return num1+num2;
    }
}
```

Output:

A screenshot of a terminal window showing the execution of a Java program. The first line shows the command prompt: "C:\Program Files\Java\jdk-20\bin\java.exe". The program then prompts for two integers. The user enters 10 for num1 and 20 for num2. The program outputs the sum: "Sum of 10 and 20 is 30". Finally, it shows "Process finished with exit code 0".

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a value of num1: 10
Enter a value of num2: 20
Sum of 10 and 20 is 30

Process finished with exit code 0
```

Problem 2: Simple Age Checker (Using a Method)

Source code:

```
package LB_Assingment3;

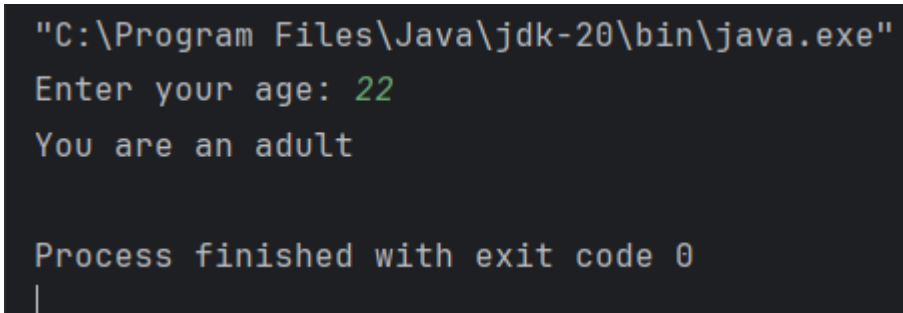
import java.util.Scanner;

public class Age_checker {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age=sc.nextInt();

        checkAgeCategory(age);
    }

    public static void checkAgeCategory(int age){
        if(age<18){
            System.out.println("You are a minor.");
        }else if(age<60){
            System.out.println("You are an adult");
        }else{
            System.out.println("You are a Senior Citizen");
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter your age: 22
You are an adult

Process finished with exit code 0
|
```

Problem 3: Print Even Numbers (Using while Loop)

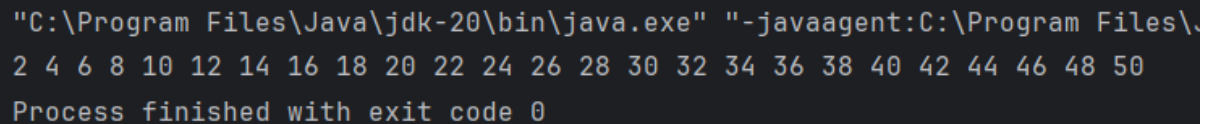
Write a Java program that prints all even numbers between 1 and 50 using a while loop.

Source code:

```
package LB_Assingment3;

public class Even_num_using_while {
    public static void main(String[] args) {
        printEvenNumber();
    }
    public static void printEvenNumber(){
        int num=1;
        while (num<=50){
            if(num%2==0){
                System.out.print(num+" ");
            }
            num++;
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe" "-javaagent:C:\Program Files\
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
Process finished with exit code 0
```

Problem 4: User Input for Positive Numbers (Using do-while Loop)

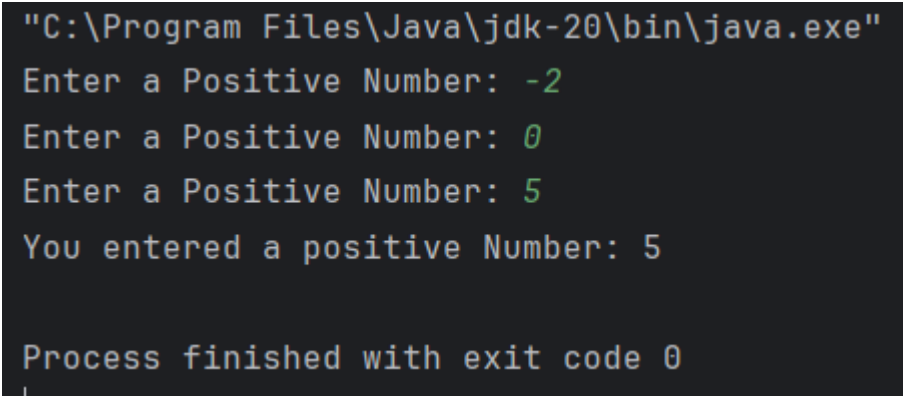
Source code:

```
package LB_Assingment3;

import java.util.Scanner;

public class Positive_num_using_dowhile {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        askForPositiveNumber(sc);
    }
    public static void askForPositiveNumber(Scanner sc){
        int num;
        do{
            System.out.print("Enter a Positive Number: ");
            num=sc.nextInt();
        }while (num<=0);
        System.out.println("You entered a positive Number: "+num);
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Positive Number: -2
Enter a Positive Number: 0
Enter a Positive Number: 5
You entered a positive Number: 5

Process finished with exit code 0
```

Problem 5: Print Multiplication Table (Using for Loop)

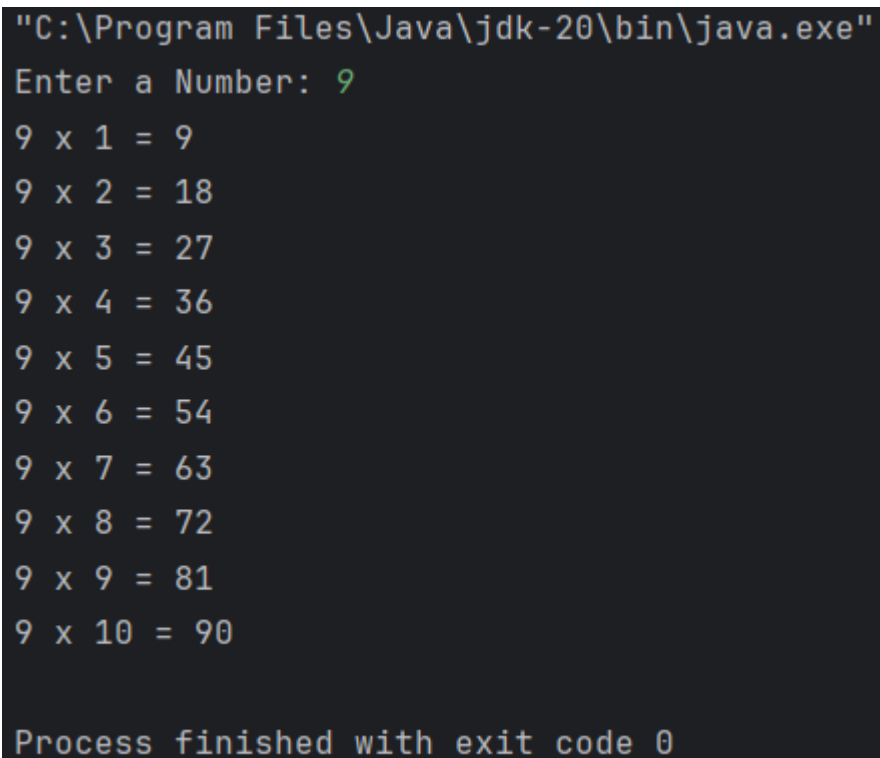
Source code:

```
package LB_Assingment3;

import java.util.Scanner;

public class Mul_Table_Using_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num= sc.nextInt();
        printMultiplicationTable(num);
    }
    public static void printMultiplicationTable(int num){
        for(int i=1;i<=10;i++){
            int mul=num*i;
            System.out.println(num+" x "+i+" = "+mul);
        }
    }
}
```

Output:



```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90

Process finished with exit code 0
```

Problem 6: Calculate the Sum of Numbers from 1 to N (Using for Loop)

Source code:

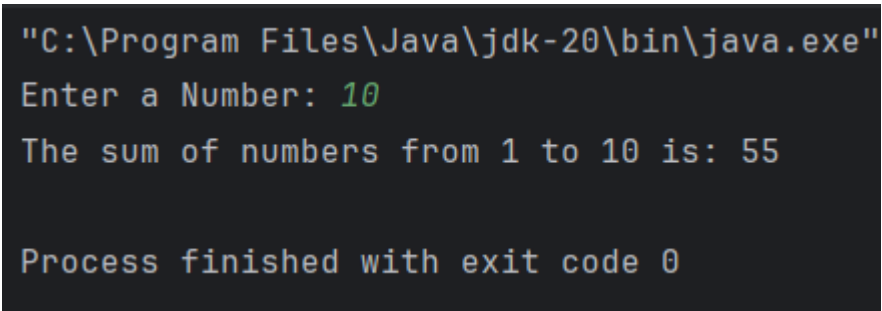
```
package LB_Assingment3;

import java.util.Scanner;

public class SON_1ton_using_forloop {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter a Number: ");
        int num= sc.nextInt();

        calculateSum(num);
    }
    public static void calculateSum(int num){
        int add=0;
        for(int i=1;i<=num;i++){
            add=add+i;
        }
        System.out.println("The sum of numbers from 1 to "+num+" is: "+add);
    }
}
```

Output:

A screenshot of a Java command prompt window. The text displayed is: "C:\Program Files\Java\jdk-20\bin\java.exe" on the first line, "Enter a Number: 10" on the second line (where '10' is green), "The sum of numbers from 1 to 10 is: 55" on the third line, and "Process finished with exit code 0" on the fourth line.

```
"C:\Program Files\Java\jdk-20\bin\java.exe"
Enter a Number: 10
The sum of numbers from 1 to 10 is: 55

Process finished with exit code 0
```

Bonus Problem: Menu-Driven Java Program (Switch-Case)

Problem 1: Grade Evaluation System

Problem 2: Leap Year Check

Problem 3: Day of the week

Problem 4: Identify Default Values of Variables

Case 5: Exit

Source code:

```
package LB_Assingment3;

import java.util.Scanner;

public class Menu_Driven {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        System.out.println("1. Grade Evaluation System");
        System.out.println("2. Leap Year Check");
        System.out.println("3. Day of the Week");
        System.out.println("4. identify the default values of Variables");

        System.out.print("Enter a number from 1 to 4: ");
        int num=sc.nextInt();

        switch (num){
            case 1:
                System.out.print("Enter marks for Maths: ");
                int maths= sc.nextInt();
                System.out.print("Enter marks for Science: ");
                int Science= sc.nextInt();
                System.out.print("Enter marks for History: ");
                int History= sc.nextInt();
                gradeEvaluate(maths,Science,History);
                break;

            case 2:
                System.out.println("Enter a year: ");
                int year=sc.nextInt();
                yearCheck(year);
                break;

            case 3:
                System.out.println("Enter a Day Number: ");
                int day=sc.nextInt();
                Day_of_week(day);
```

```

        break;

    case 4:
        Uninitialized_var();
        break;

    default:
        System.out.println("Exit");
    }
}

public static void gradeEvaluate(int maths,int Science,int History){
    int avg=(maths+Science+History)/3;
    if(avg>=90){
        System.out.println("Average Marks: "+avg);
        System.out.println("Grade: A");
    }else if(avg<=89 && avg>=70){
        System.out.println("Average Marks: "+avg);
        System.out.println("Grade: B");
    }else if(avg<=69 && avg>=50){
        System.out.println("Average Marks: "+avg);
        System.out.println("Grade: C");
    }else if(avg<=49 && avg>=30){
        System.out.println("Average Marks: "+avg);
        System.out.println("Grade: D");
    }else{
        System.out.println("Average Marks: "+avg);
        System.out.println("Fail");
    }
}

public static void yearCheck(int year){
    if((year%4==0 && year%100!=0)|| year%400==0){
        System.out.println(year+" is a leap year");
    }else{
        System.out.println(year+" is not a leap year");
    }
}

public static void Day_of_week(int day){
    switch (day){
        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:

```



```

        System.out.println("Wednesday");
        break;
    case 4:
        System.out.println("Thursday");
        break;
    case 5:
        System.out.println("Friday");
        break;
    case 6:
        System.out.println("Saturday");
        break;
    case 7:
        System.out.println("Sunday");
        break;
    default:
        System.out.println("Invalid day Number");
    }
}

```

```

public static void Uninitialized_var(){
    byte a = 10;
    short b = 1000;
    int c = 100000;
    long d = 1000000000000L;
    float e = 10.5f;
    double f = 20.55;
    char g = 'P';
    boolean h = true;

    System.out.println("byte: " + a);
    System.out.println("short: " + b);
    System.out.println("int: " + c);
    System.out.println("long: " + d);
    System.out.println("float: " + e);
    System.out.println("double: " + f);
    System.out.println("char: " + g);
    System.out.println("boolean: " + h);
}
}

```